QUANSER
INNOVATE · EDUCATE

Concept Review

Filtering

Why Do We Need Filtering?

Ever listened to a song and tried to focus on one specific instrument? In order to do this your brain has to "filter" out all of the other instruments, or what is referred to in signal processing as "noise". In a similar way, data read from sensors may have noise due to the environment they are placed in or the inherent design of the sensor. Filtering is essential because it allows for the removal of noise to isolate a specific signal of interest. There are many different filtering methods that exist to remove noise from a signal.

Introduction

Filtering is a step in signal processing that reduces unwanted components from a signal. Sensors are often subject to noise, interference, and artifacts, which can contaminate the data and make it less reliable for analysis or control. For example, in medicine, if you wanted to measure someones ECG (heart's electronic signals), there are several noises that would need to be filtered out, including electrode motion artifacts (moving), myoelectric control (electrical activity of muscles), and powerline interference. When measuring signals from motors, vibrations from the spinning motors and propellers will add noise in the accelerometer and gyroscope measurement data. Filters allow for removal of noise above or below a threshold frequency, or a specific frequency to isolate a specific signal of interest. This document will cover the four major filters: low-pass, high-pass, band-pass and notch filters.

Filters are often characterized using a transfer function in the s-domain (Laplace domain), in terms of the complex frequency variable s. The filter is visualized on a Bode plot, which displays the response of a system as a function of the input frequency. Bode magnitude plots show how much a signal is amplified or attenuated (reduced) at each frequency in decibels.

Low-Pass Filters

Low-pass filters are one of the most commonly used filters. As the name implies, this filter aims to allow noise of frequencies below a certain threshold to pass through, while attenuating frequencies *above* the threshold. This threshold, ω_{co}, in radians/s, is called the cut-off frequency of the filter. This behaviour is best described by the filter bode plot (Figure 1), which shows low-pass filters ω_c for 10 rad/s, 100 rad/s and 1000 rad/s. All higher frequency components of the signal get attenuated by at least -3dB or around 50%.

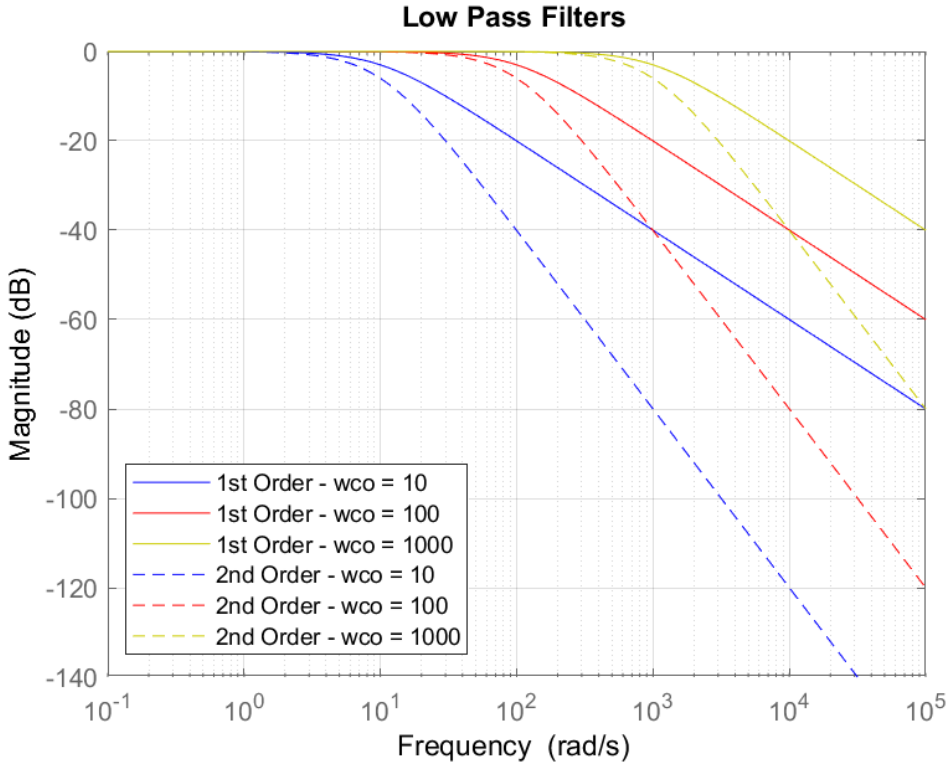

Figure 1: Bode response plot for low-pass filters of various cut-off frequencies

As seen, the magnitude of the signal above the cutoff frequency in each case rapidly decreases at a linear rate. Note that a drop of X dB corresponds to a magnitude reduction by $10^{X/10}$. The magnitude of signal noise below the cutoff frequency is left untouched. The second order filter can be seen to produce effects that are somewhat similar to a first-order low-pass filter. However, it attenuates unwanted frequencies at a faster rate.

The first order transfer function for this filter takes the following form:

$$H_{LPF}(s) = \frac{\omega_{co}}{s + \omega_{co}} \tag{1}$$

The second order transfer function for this filter takes the following form:

$$H_{LPF}(s) = \frac{\omega_{co}^2}{s^2 + 2\omega_{co}s + \omega_{co}} \tag{2}$$

Here, ω_{co} represents the cut-off frequency.

High-Pass Filters

The high-pass filter aims to allow noise of frequencies above the cut-off frequency to pass through, while attenuating frequencies *below* the cut-off. This can be useful for removing DC offset or drift, or to monitor sudden disturbances in a system by isolating high-frequency components. The behaviour of a high-pass filter is illustrated by the filter bode plot (figure 2), which shows three 1st order and three 2nd order high-pass filters with ω_{co} being 10 rad/s, 100 rad/s and 1000 rad/s.

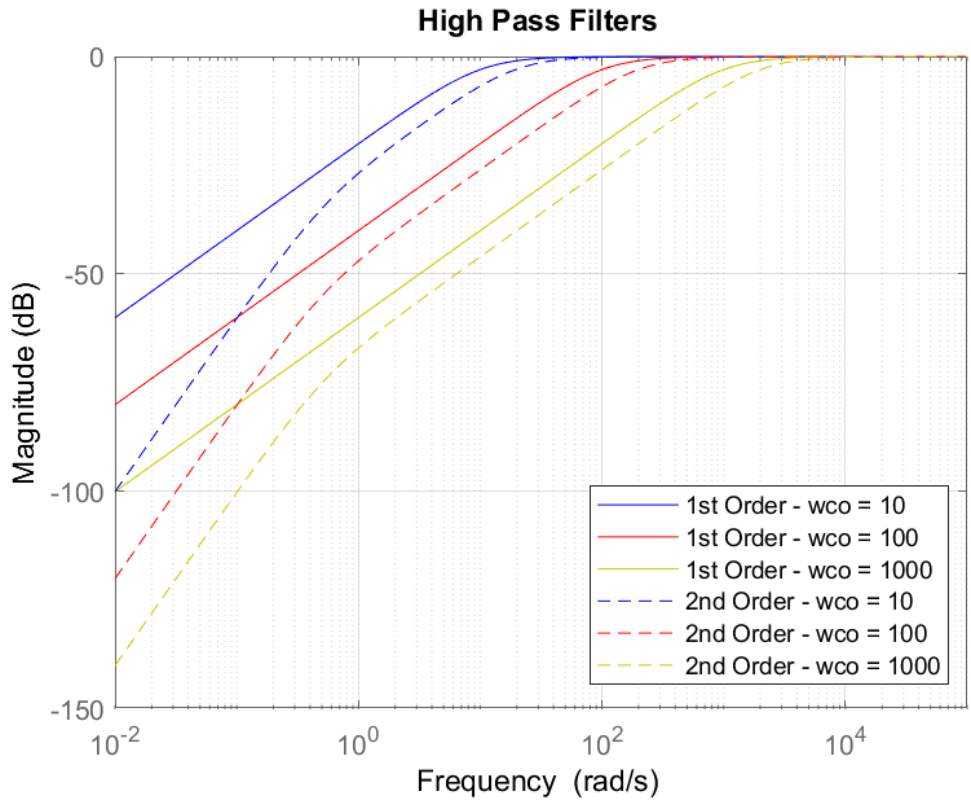

Figure 2: Bode response plot for 1st and 2nd Order high-pass filters of various cut-off frequencies

As you can see, the magnitude of signal noise rapidly increases at a linear rate up to the cutoff frequency. This shows that below the cutoff, the signal is being reduced (filtered). The magnitude of signal noise above the cutoff frequency is left untouched (0 dB). The second order filter can be seen to produce effects that are somewhat similar to a first-order high-pass filter. However, just like with low-pass filters, it attenuates unwanted frequencies at a faster rate. The transfer function of this filter looks like:

$$H_{HPF}(s) = \frac{s}{s + \omega_{co}} \tag{3}$$

The second order transfer function for this filter looks like:

$$H_{HPF}(s) = \frac{s^2}{s^2 + 2\omega_{co}s + \omega_{co}} \tag{4}$$

Here, ω_{co} represents the cut-off frequency.

Band-Pass Filters

A band-pass filter allows only a specified range of frequencies, blocking frequencies above or below the range. A band-pass filter can be thought of as a low and high pass filter. This behaviour is best described by the filter bode plot (figure 3), which shows band-pass filters with ω_0 being 10 rad/s, 100 rad/s and 1000 rad/s.

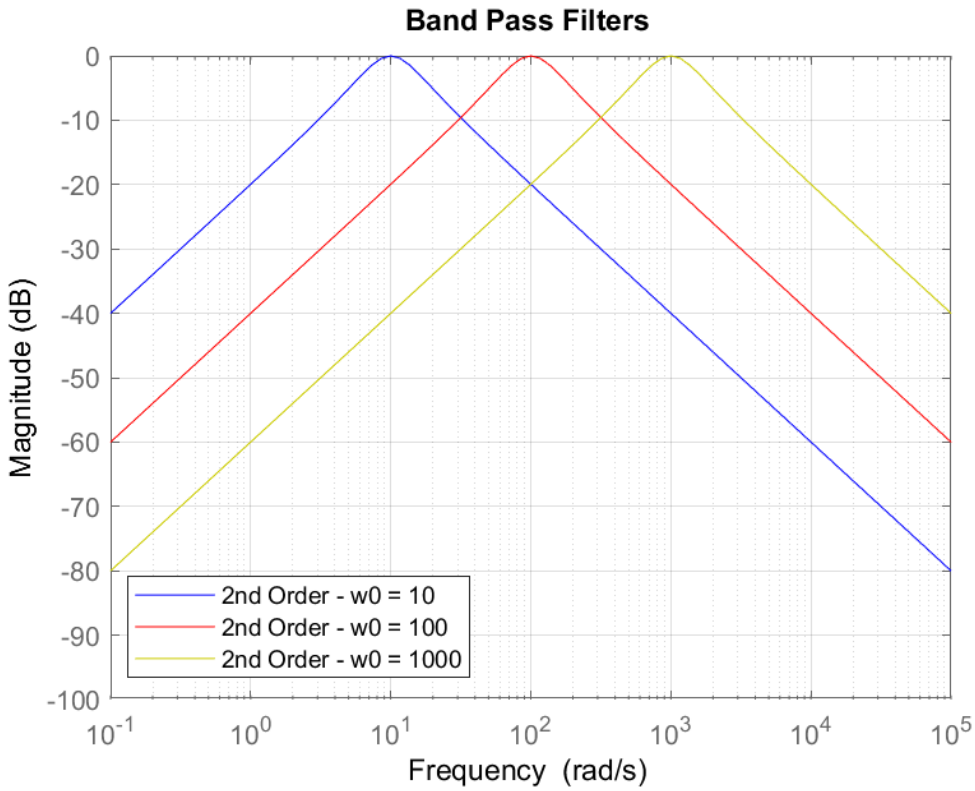

Figure 3: Bode response plot for band-pass filters of various cut-off frequencies

As you can see, the magnitude of the signal rapidly increases at a linear rate up to ½ the bandwidth specified from the center frequency chosen, then rapidly decreases at a linear rate ½ the bandwidth away from the center frequency chosen. This shows that below the cutoff, the signal is being reduced (filtered). The magnitude of signal noise between the two cutoff frequencies (within the bandwidth) is left untouched (0 dB).

The transfer function for a second-order band-pass filter is the following:

$$H_{BPF}(s) = \frac{\frac{\omega_0}{\beta} s}{s^2 + \frac{\omega_0}{\beta} s + \omega_0^2} \tag{3}$$

Here, ω_0 represents the center frequency and β is the desired bandwidth.

In practice, no band-pass filter is ideal (or perfect). The filter will not attenuate all frequencies outside the desired frequency range completely. There is typically a region just outside the desired range where frequencies are attenuated, but not removed completely. This is known as filter roll-off and is usually expressed in dB of attenuation per octave or decade of frequency. When designing filters, the goal is typically to make the roll-off as narrow as possible around the desired signal frequency.

Notch Filters

Unlike the band-pass filters which filter everything *outside* of a band width, notch filters, also known as band stop filters or band rejection filters, attenuate (reduce or remove) all signals *within* a specified band width. This band width is called the stop band. This behaviour is best described by the filter bode plot (figure 4), which shows notch filters with ω_0 being 10 rad/s, 100 rad/s and 1000 rad/s and a bandwidth of 2 Hz.

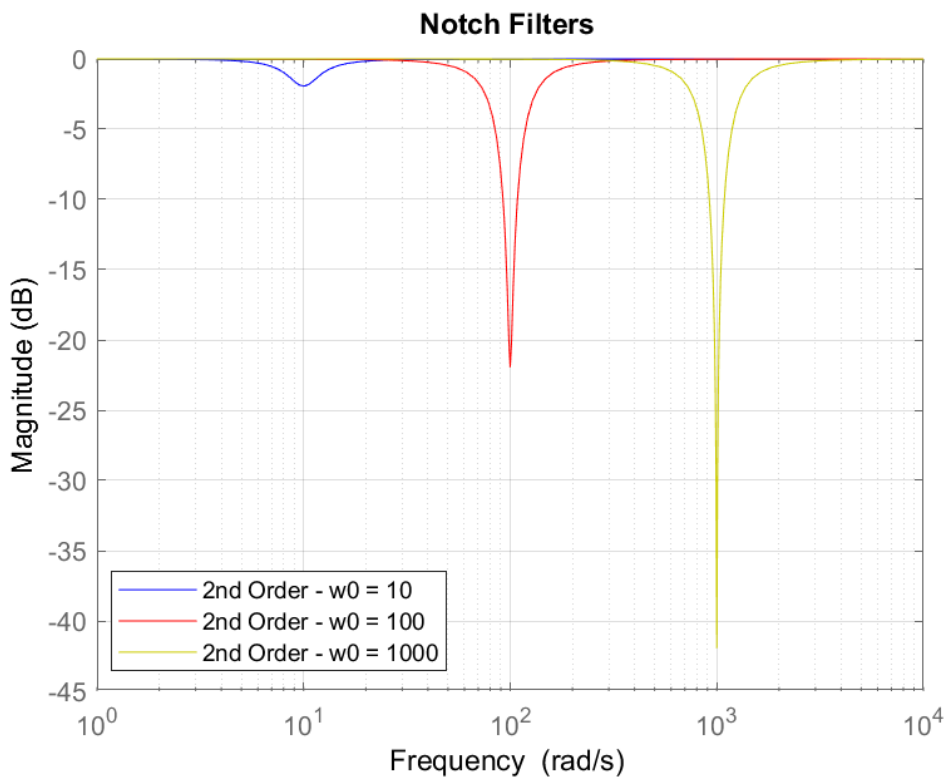

Figure 4: Bode response plot for notch filters of various cut-off frequencies at a bandwidth of 2 Hz.

Since there is two frequencies needed in order to calculate the The transfer function for a second-order notch filter is the following:

$$H_{NF}(s) = \frac{s^2 + 2\beta s + \omega_0^2}{s^2 + \frac{\omega_0}{\beta} s + \omega_0^2} \tag{4}$$

Here, ω_0 represents the center frequency and β is the bandwidth desired to be removed.

Complementary Filters

Combining our understanding of low-pass and high-pass filters we can now take a look at complementary filters.

If a noisy signal was sent through a high-pass filter and a low-pass filter with the same cut-off frequency in parallel, and the filtered signals were added together, one would expect to achieve the same signal again. This can be illustrated through the following equation

$$C_{filtered} = H(s)x + G(s)x = \left(\frac{s}{s+\omega_n}\right)x + \left(\frac{\omega_n}{s+\omega_n}\right)x = x$$

The high-pass and low-pass filters with the same cut-off frequency are complementary in nature, a fact that makes complementary filters very useful in fusing sensor data. For more information check out the Concept Review - Sensor Fusion.

© Quanser Inc., All rights reserved.

QUANSER
INNOVATE · EDUCATE

Solutions for teaching and research. Made in Canada.